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Potential is “intrinsic energy of inertia” [label](#)

$$\nabla \cdot \epsilon_0 \mathcal{E} = \varrho \quad \nabla \cdot \mathcal{B} = 0 \quad \nabla \times \mathcal{E} = -\frac{\partial}{\partial t} \mathcal{B} \quad \nabla \times \frac{1}{\mu_0} \mathcal{B} = j + \epsilon_0 \frac{\partial}{\partial t} \mathcal{E} \mathcal{B} = \nabla \times U_m \quad \mathcal{E} = -\nabla U_e - \frac{\partial}{\partial t} U_m \quad (1)$$

$\Rightarrow$ Wave equation for potential:

$$\Delta \cdot U - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} U = J \quad (2)$$

$\Rightarrow$ Potential by “ignition J” (cause, inhomogeneity) from source at rest, $v = 0$:

$$U = -\frac{1}{4\pi} \int \frac{J}{|r|} dr' \quad (3)$$

elec. Potential structure: $J = -\frac{1}{\epsilon_0} \varrho$

$$U_e := \Phi \stackrel{\text{elec. charge dens.}=\varrho}{=} \frac{1}{4\pi} \frac{1}{\epsilon_0} \int \frac{\varrho}{|r|} dr' \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\epsilon_0} Q = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\epsilon_0} It \quad (4)$$

magn. Potential structure: $J = -\mu_0 j$

$$U_m := A \stackrel{\text{elec. current dens.}=j}{=} \frac{\mu_0}{4\pi} \int \frac{j}{|r|} dr' \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \mu_0 I = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{c^2 \epsilon_0} I = \frac{1}{c^2} \frac{1}{t} U_e \stackrel{\text{source in uniformal movement, } v=\text{const.}}{=} \quad (5)$$

Electric endogen energy $E_{pot} = mc^2$:

$$e = \text{electron charge} \quad q = ne = \text{charge} \quad n \in \mathbb{N} \quad (6)$$

$$\text{Potential} \quad U = \frac{E_{pot}}{q} = \frac{(mc^2)^2}{ne} = \frac{m}{q} mc^4 = \kappa mc^4 \quad (7)$$

Electric induced energy (exogenous excitation) $E_{kin} = \frac{1}{2} mv^2 \hat{=} (pc)^2$:

E_{kin} Energy density electric:

$$w_e = \frac{1}{2} \epsilon_0 \mathcal{E}^2 \quad (8)$$

E_{kin} Energy density magnetic:

$$w_m = \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2 \quad (9)$$

E_{kin} Electric Energy:

$$W_e = \frac{1}{2} C U^2 \quad (10)$$

E_{kin} Magnetic Energy:

$$W_m = \frac{1}{2} L I^2 \quad (11)$$

Power:

$$P = VI = (U_1 - U_2) \frac{Q}{t} = \frac{E_1 - E_2}{t} = \frac{W}{t} = \frac{Fr}{t} = Fv \quad (12)$$

Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \epsilon_0 \quad (13)$$

Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c} \frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \quad (14)$$

Energy conservation for (large extended) conductive medium:

$$\int \left(\frac{\partial}{\partial t} (we + w_m) + \cdot \mathcal{S} + j \cdot \mathcal{E} \right) dr = \frac{d}{dt} W_e + \frac{d}{dt} W_m + D_S + \int (j \cdot \mathcal{E}) dr \stackrel{\text{low frequency}}{=} \frac{1}{2} \frac{d}{dt} (L I^2) + \frac{1}{2} \frac{d}{dt} \left(\frac{Q^2}{C} \right) \quad (15)$$

Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potential difference, tension):

$$V = U_1 - U_2 = r\mathcal{E} = \dot{\Psi} + RI + \frac{I}{C}t = L\dot{I} + RI + \frac{Q}{C} \quad (16)$$

$$\Delta \cdot := \text{div grad} = \text{Laplacian}$$

J = "ignition of inhomogeneity, cause" j = Current density ϱ = Charge density

V = Potential difference I = Current R = Resistance

L = Inductance C = Capacitance Q = charge (ion) $\Psi = LI$ = mag. Flux

$\mathcal{S} = \mathcal{E} \times \mathcal{H}$ = Poynting-Vector = radiation intensity

$D_S = \int (\cdot \mathcal{S}) dr = \int \text{div}(\mathcal{E} \times \mathcal{H}) dr$ = dissipation energy lost from radiation ≈ 0 for low frequencies ω

$d = \sqrt{\frac{1}{\mu\sigma} \frac{2}{\omega}}$ = skin depth (surface) of conductor $r = \frac{d}{\sqrt{2}i} \rho = \frac{d}{1+i} \rho$ = radius (extension) of conductor

$\rho = kr$ = "radius of curvature k "

$\omega = 2\pi\nu = k\lambda \frac{1}{T} = \sqrt{\frac{k}{m}} = \frac{v}{r} \hat{=} kc \hat{=} \frac{1}{\sqrt{LC}} \hat{=} \sqrt{\frac{g}{r}} = \text{frequency}$

μ = permeability of conductor σ = conductivity of conductor (dielectricum)

Low Frequency:

$$\begin{aligned} \omega &\ll \frac{1}{\mu\sigma} \frac{2}{r^2} \iff r \ll d \iff \rho \ll 1 \\ L &\approx \text{const.} \quad C \approx \text{const.} \quad D_S \approx 0 \end{aligned} \tag{17}$$

High Frequency:

$$\begin{aligned} \omega &\gg \frac{1}{\mu\sigma} \frac{2}{r^2} \iff r \gg d \iff \rho \gg 1 \\ L(t) &\neq \text{const.} \quad C(t) \neq \text{const.} \quad \text{for Dipol: } D_S \propto \omega^4 \gg 0 \end{aligned} \tag{18}$$

Energy:

Action Quantum, Planck Constant (Energy-Frequency-Slope)¹:

$$h = \tan(\alpha) = \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \frac{E_1 - E_2}{\nu_0} = \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k \lambda \quad (19)$$

$$\hbar = \frac{E_1 - E_2}{\omega} = \quad (20)$$

$$E_{kin}^{max} = eV_0 = e \frac{Q}{C} = h(\nu - \nu_0) \quad (21)$$

V_0 = Opposing Potential ($I = 0$)

$\alpha = \angle(E_{kin}, \nu)$ e = electron charge Q = ion charge

$$C = \text{Capacitance} \quad \nu = \frac{1}{2\pi} \frac{1}{\sqrt{LC}}$$

$$\Rightarrow \text{Inductance } L = \frac{4\pi^2}{C} \left(\frac{e}{h} Q + \nu_0 \right)^2$$

$\Rightarrow$ Ionisation Energy of chemical element $E_i = h\nu = h \frac{\lambda}{c} = \hbar \omega = \frac{h}{k\lambda} \omega$

$\Rightarrow$ Energy released (gained) $E_{gain} = E_{pot} - E_i = mc^2 - h\nu$

¹Wirkungsquantum